Power supply: 19V –> 5V (AP63205) with USB OR-ing, 5V –> 3.3V (AP63203)

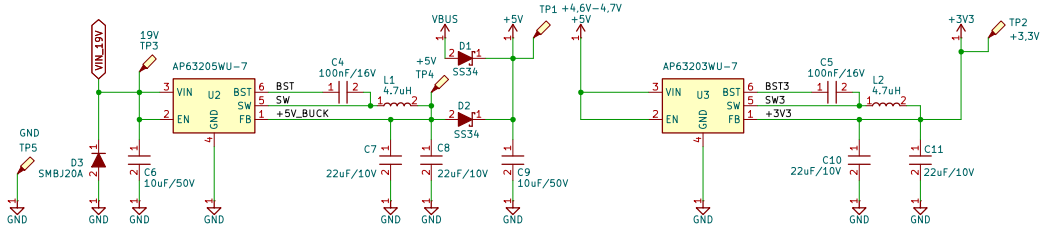
AP63205 (19V –> +5V_BUCK) and USB VBUS are OR-ed with two Schottky diodes into the +5V rail.

+5V rail is about 4.6–4.7 V after the diode drop. USB-only operation is limited to 500 mA total.

AP63203 runs from the +5V rail, so 3.3V is available from 19V input or from USB alone.

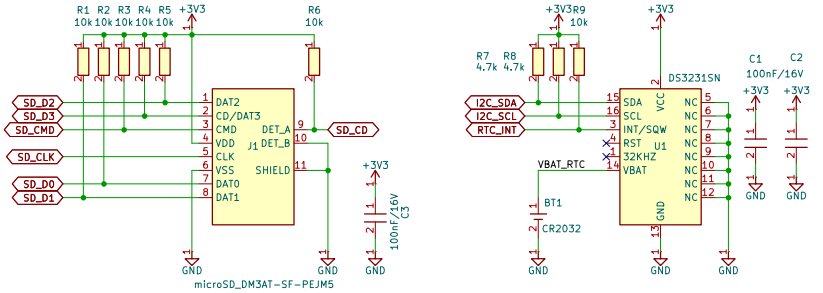
Components per Diodes Inc. datasheet DS41326 (tables 2 and 3): 10uF in, 2x22uF out, 100nF boot.

Datasheet suggests 3.9uH for the 3.3V part; 4.7uH is within the allowed range (2.2–10uH).



+5V, +3V3, GND, VBUS are global power nets.

microSD card and real-time clock (DS3231 with coin cell backup)



SD card in 4-bit mode: CLK=IO6, CMD=IO7, D0=IO5, D1=IO4, D2=IO18, D3=IO17. Card detect=IO13.

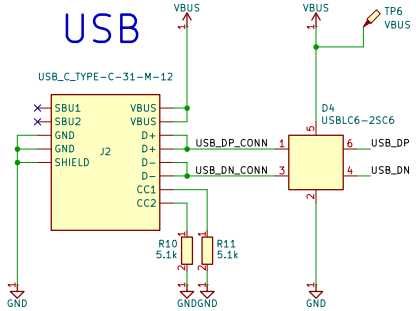
Pull-ups on all SD lines keep the card in SD mode. Pads 9–11 of the socket (detect/shield) must be verified.

Card detect switch is normally open (closes when a card is inserted): SD_CD reads low with card present.

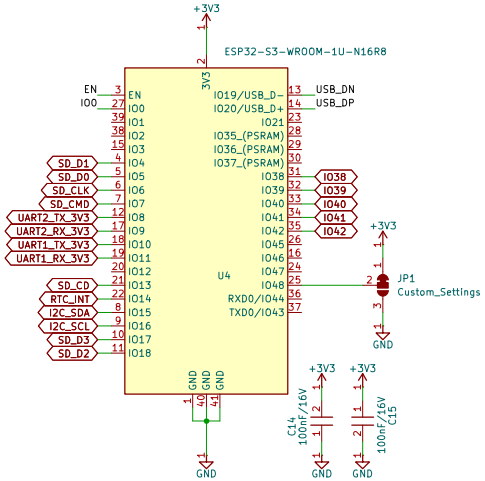
DS3231: INT/SQW (open-drain, 10k pull-up) goes to IO14 as interrupt / deep-sleep wake source.
RST and 32KHZ unused. N.C. pins 5–12 tied to GND.

Coin cell: non-rechargeable CR2032 connected directly to VBAT (2.3–5.5 V).

USB



EBC Tester – ESP32-S3 core circuit (MCU, USB, reset/boot)



WROOM-1U: external 2.4 GHz antenna (u.FL connector) required

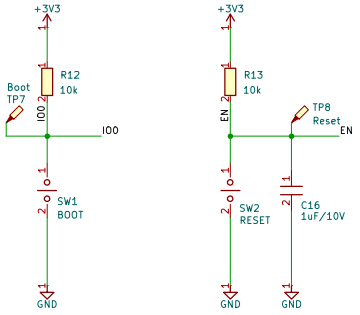
IO35/IO36/IO37 are used internally by the octal PSRAM (N16R8) – do not use

UART1 (Tester): TX=IO10, RX=IO11 | UART2 (Dongle): TX=IO8, RX=IO9

SD card: CLK=IO6, CMD=IO7, D0=IO5, D1=IO4, D2=IO18, D3=IO17,
CD=IO13 (CD is a proposal)

I2C (RTC): SDA=IO15, SCL=IO16 | RTC interrupt INT/SQW=IO14
(RTC GPIO, deep-sleep wake capable)

Boot and Reset switch



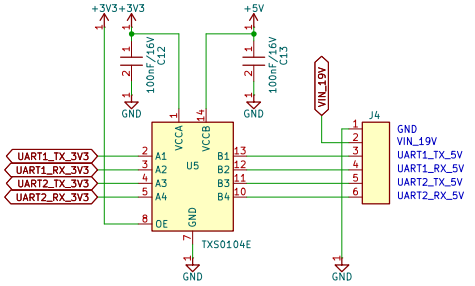
Native USB on IO19/IO20. Download mode: hold BOOT, press RESET.

UART connector

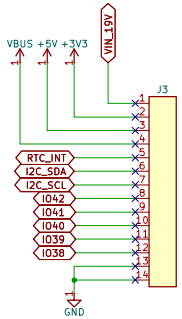
Connector pin order: 1=GND, 2=+19V, 3=UART1_TX, 4=UART1_RX, 5=UART2_TX, 6=UART2_RX.

UART1 = Tester, UART2 = Dongle. TX/RX names are from the ESP32 point of view.

Connector pitch 2.5 mm XH-style. +5V rail is about 4.6–4.7 V.



Expansion connector



(C) 2026 by Thomas Stoll
Bauan / Batangas
Philippines
Anthracite

Sheet: /
File: EBCTester.kicad_sch

Title: EBCTester

Size: A3	Date: 2026-10-30	**Rev: 1**
KiCad E.D.A. 10.0.4		Id: 1/1